AI For Engineers-Set B

shubhashri.gadewar24@vit.edu Switch account

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Email *

shubhashri.gadewar24@vit.edu

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Shubhashri Kiran Gadewar

College Name-MANDATORY FOR **ALL** COLLEGE STUDENTS (**Please select your specific college name carefully and accurately**) *

Vishwakarma Institute of Technology

Year-MANDATORY FOR **ALL** COLLEGE STUDENTS *

Year II

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12

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CS-D

A student asks AI, 'Design a manufacturing monitoring system.' The answer is generic. What change would most improve the quality of the response? * 1 point

- ( ) Ask for a shorter answer so AI gets to the point faster
- (•) Add context, constraints, success metrics, and likely failure modes
- ( ) Switch to a different AI model without changing the prompt
- ( ) Remove technical details so the AI can be more creative

Which best captures the main difference between using AI to finish work and using AI to understand work? * 1 point

- ( ) Finishing work is faster, while understanding work is usually impossible with AI
- (•) Finishing work focuses on output completion, while understanding work focuses on deeper reasoning and mental models
- ( ) Finishing work is for beginners, while understanding work is only for researchers
- ( ) There is no meaningful difference if the answer looks correct

[Multi-select] Which prompt elements were emphasized in the talk as especially important when talking to AI like an engineer? * 1 point

- [x] Context
- [x] Constraints
- [x] Failure modes
- [x] Trade-offs

An engineer gets a vague AI answer to a design problem. According to the talk, what is the most likely reason? * 1 point

- ○ AI tools are mainly built for non-technical users
- ○ The engineer used the wrong browser
- ◉ The prompt lacked clear structure and boundaries
- ○ The problem was too small for AI to handle

A plant supervisor sees defect rates rise only on one shift and only for one machine family. Which habit from the talk is being applied best? * 1 point

- ○ Jump straight to the most expensive fix
- ◉ Frame the problem carefully before solving it
- ○ Ask AI for the final answer without local data
- ○ Ignore narrow patterns and focus only on company-wide trends

[Multi-select] A senior engineer is troubleshooting an intermittent sensor failure. Which actions fit the hypothesis-based troubleshooting pattern from the talk? * 1 point

- [x] List plausible causes first
- [x] Rank causes by likelihood
- [x] Ask for diagnostics to test each cause
- [ ] Choose the first explanation that seems possible and stop

When the speaker said most students use AI like a search engine while professionals use it like a thinking partner, what distinction was he highlighting? * 1 point

- ( ) Professionals prefer longer answers
- (•) Professionals use AI for structured reasoning, not just information lookup
- ( ) Students should avoid AI entirely
- ( ) Professionals never make mistakes with AI

A student asks AI to explain a control-system concept and also asks for physical intuition, common failure cases, and when the method should not be used. What is this an example of? * 1 point

- ( ) Prompting for memorization only
- ( ) Using AI as a calculator
- (•) Using AI to deepen conceptual understanding
- ( ) Avoiding trade-offs

In the talk's PID controller example, what was the broader lesson? * 1 point

- ( ) Only advanced control experts should use AI
- (•) AI can help move from exam-ready definitions to industry-ready understanding
- ( ) PID is always better than model predictive control
- ( ) Derivative control should always be removed

In the talk, why did the structured Laplace transform prompt produce a much better answer than the weak prompt? * 1 point

- ( ) The structured prompt used more difficult vocabulary
- (•) The structured prompt added learner context, use-case relevance, and conceptual goals
- ( ) The AI had warmed up after the first attempt
- ( ) Laplace transform is a topic AI already memorizes perfectly

A mechanical student asks AI, 'Explain bearings.' A stronger version would be: * 1 point

- ( ) Explain bearings in one sentence
- ( ) Explain bearings as if I know nothing and avoid all technical details
- (•) I am a diploma student who understands friction and shafts; explain bearing selection with intuition, load trade-offs, failure modes, and one real machine example
- ( ) Give me the top 50 facts about bearings

Why did the talk recommend ending prompts with 'critique your answer'? * 1 point

- ( ) It makes the answer shorter
- (•) It forces AI to reflect on weaknesses, assumptions, and possible blind spots
- ( ) It guarantees the answer is correct
- ( ) It stops AI from making trade-offs

[Multi-select] Which situations from the spirit of the talk most clearly require human validation even if the AI output sounds confident? * 1 point

- [x] A medical-device control setting recommendation
- [x] A structural load estimate for a real installation
- [ ] A student asking for a motivational summary of a chapter
- [x] A firewall rule set being proposed for a production network

Which statement best reflects the AI Amplification Rule from the talk? * 1 point

- ( ) AI mainly replaces the need for thinking
- ( ) AI gives strong answers even to weak prompts
- (•) AI amplifies the quality of the thinking you bring into the interaction
- ( ) AI should only be used after solving the problem manually

A student uses AI to generate a circuit explanation, then checks assumptions, asks follow-up questions, and tests whether the logic fits the real setup. Which career skill is this building most directly? * 1 point

- (•) Validation
- ( ) Memorization
- ( ) Presentation design
- ( ) Typing speed

In the scaling architecture demo from the talk, what was the central engineering skill being demonstrated? * 1 point

- ○ Debugging syntax errors
- ◉ Making trade-off based strategic decisions under constraints
- ○ Memorizing cloud service names
- ○ Avoiding all operational complexity

In the talk's server scaling example, which option was described as simpler initially but limited by a ceiling and single-machine dependence? * 1 point

- ○ Horizontal scaling
- ○ Edge-only scaling
- ◉ Vertical scaling
- ○ Serverless scaling

A startup expects unpredictable traffic spikes but has very limited DevOps expertise. Which answer best matches the kind of judgment the talk encouraged? * 1 point

- ○ Choose whichever option sounds more modern
- ○ Recommend based only on cost this month
- ◉ Evaluate trade-offs: short-term simplicity versus future resilience and scaling needs
- ○ Avoid making a recommendation until traffic becomes impossible to manage

When average output quality rises because AI can produce decent baseline work, what happens to 'average' performance according to the talk? * 1 point

- ( ) It becomes more valuable because more people can do it
- ( ) It stays equally valuable if formatted well
- (•) It becomes less differentiated and easier to automate
- ( ) It becomes the main basis for hiring

[Multi-select] According to the talk, which capabilities will still matter strongly in an AI-rich career environment? * 1 point

- [x] Judgment
- [x] System integration
- [x] Ownership
- [x] Validation

A student copy-pastes AI-generated code through college but cannot explain design choices during an interview. Which warning from the talk does this illustrate most directly? * 1 point

- ( ) AI tools are unreliable for code
- (•) Using AI as a shortcut can reduce depth and real problem-solving ability
- ( ) Interviews should be removed from hiring
- ( ) Students should never automate any repetitive task

A hiring team uses AI to screen resumes based only on past successful hires. What is the main ethical risk here? * 1 point

- (●) The system may reinforce existing bias patterns from historical decisions
- ( ) The system will automatically become illegal in every country
- ( ) The AI will always reject more candidates than humans
- ( ) Bias is not relevant if accuracy looks high

[Multi-select] Which practices help reduce bias and overtrust when using AI for decision support? * 1 point

- [x] Check whether important groups or scenarios are underrepresented
- [x] Ask what assumptions the model is making
- [x] Validate outputs against real-world evidence
- [ ] Assume confident wording means low risk

An AI tool suggests a highly polished answer for an education-policy question, but provides no sources and ignores local context. What should a careful student do? * 1 point

- ( ) Use it because polished language usually indicates strong reasoning
- (●) Treat it as a draft, then verify evidence, assumptions, and context
- ( ) Reject AI entirely for all policy topics
- ( ) Ask the AI to make the answer more confident

A robotics student asks AI for a troubleshooting plan and receives ten possible causes. What is the best next step based on the talk? * 1 point

- ○ Test all ten randomly
- ○ Pick the most technical-sounding cause
- ◉ Organize the causes into categories and prioritize diagnostics
- ○ Stop there because the hard work is done

Which prompt is most aligned with the talk's engineering template? * 1 point

- ○ Explain this topic quickly
- ◉ Here is my situation, constraints, required output format, trade-offs to consider, failure modes, and then critique your answer
- ○ Give me the best answer and do not mention uncertainty
- ○ Summarize everything in motivational language only

A student wants AI help on a mechatronics project. Which request is likely to produce the most useful answer? * 1 point

- ○ Tell me everything about mechatronics
- ○ Give me the final design with no questions asked
- ◉ Given this actuator load, budget, safety limit, and accuracy target, compare three feasible architectures and explain trade-offs
- ○ Write a long answer with advanced terminology

[Multi-select] Which of the following are signs that a student is using AI more like a thinking partner than a shortcut? * 1 point

- [x] They ask follow-up questions to test assumptions
- [x] They use AI outputs to build mental models
- [x] They validate recommendations against real constraints
- [ ] They paste answers directly without understanding them

The talk argues that intelligence is becoming programmable. In practical terms, what does that suggest for engineering students? * 1 point

- ( ) Only software students need to care about AI
- ( ) Students should focus less on thinking and more on tool collection
- (•) Students should learn how to structure problems so AI can extend their reasoning
- ( ) Engineering theory is no longer useful

Which closing message is most faithful to the talk? * 1 point

- ( ) The main goal is to find the fastest AI tool
- (•) The key question is whether you are upgrading your thinking fast enough to work effectively with AI
- ( ) AI will make engineering knowledge unnecessary
- ( ) The safest strategy is to avoid AI until industry standards settle

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